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Dirichlet prefaces and notices

Current cumulative text: LII-LIX, *Gesammelte mathematische Werke*, vol. III, pp. 392-427

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LII. Preface to the first edition of Dirichlet's *Lectures on Number Theory*. 1863

Immediately after Dirichlet's death I was repeatedly asked to publish, in as faithful a form as possible, the university lectures that he had delivered and that had contributed so extraordinarily much to spreading acquaintance with newer and more refined parts of mathematics. I believed I could comply with this request all the more readily because, in the years 1855 to 1858, I had heard the most important of these lectures in Göttingen, and because I had also had many opportunities, in personal intercourse with Dirichlet, to become acquainted with his reasons for the method of presentation that he followed. After Dirichlet's relatives also authorized me to do so, I now present to the mathematical public an elaboration of the lecture on number theory, in which the course followed by Dirichlet in the winter of 1856-1857 has in essentials been retained. He himself at that time conceived the thought of publishing these lectures; and since he had never written out his lecture, a notebook written by me, though containing only the principal points of the proofs, served him in making an approximate estimate of the extent of the individual sections. In frequently recurring conversations about this plan he expressed the intention, when publishing it, of adding several sections that could not be absent from a textbook, but that had had to be passed over in that winter lecture for lack of time. For the present publication, therefore, the notebook just mentioned has indeed essentially been taken as the basis; but partly from older notebooks, partly from Dirichlet's papers, and finally also quite according to my own judgment, I have made additions of no inconsiderable extent, which I believe I must state here in order to assume responsibility for them. They are contained in paragraphs 105 to 110, 121 to 144, and in the notes placed immediately below the text.

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It is my intention to let this first volume, whose completion has until now been delayed by other work, be followed first by a second, less extensive volume, in which the lecture on the forces acting inversely as the square of the distance is to be reproduced.

Brunswick, October 1863.

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LIII. Notice of the first edition of Dirichlet's *Lectures on Number Theory*

[*Göttingische gelehrte Anzeigen*, year 1864, pp. 121–124.]

The undersigned editor, while a Privatdozent at the University of Göttingen, attended in the winter of 1856–1857 a lecture by Dirichlet on number theory which, although it began with the elements, was devoted mainly to the theory of quadratic forms and treated it more completely than in earlier years. The brief notes written down by him each day after the lecture, which contained almost only the main points of the proofs and were of course not at all intended for publication, were looked over by Dirichlet, who at that time was considering the publication of this lecture and sought in this way to obtain an overview of the extent of its individual parts; it is known that he himself never wrote out his lectures. Since the variety of methods that serve to prove one and the same theorem forms one of the chief attractions of number theory, and since the elements in that lecture could in any case be treated only briefly, it was not Dirichlet's intention, in publishing a textbook on number theory, to restrict himself to the content of this lecture. Rather, he expressed the intention of adding various completions by which the work would take shape as a rounded whole. When the hope of possessing such a work was frustrated by Dirichlet's too early death, the undersigned undertook, after repeated requests, to reconstruct and publish Dirichlet's lectures in as faithful a form as possible, taking the notebook mentioned above as basis but with regard to completions of the kind named. The present work is the result of his work over several years.

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As concerns the outward form, it seemed necessary to facilitate the overview by division into sections and paragraphs. The first section treats divisibility, the second the congruence of numbers, the third quadratic residues; in the fourth the elements of the theory of binary quadratic forms are presented, and the fifth contains the solution, first given by Dirichlet, of the problem of determining the number of classes into which the binary quadratic forms of given determinant divide. Alongside the main lecture proper, Dirichlet gave a supplementary lecture in which several important auxiliary theorems belonging to other domains were proved. This separation has been retained so as not to interrupt the line of thought of the fifth section, which is in any case not so easy for the beginner to grasp; the content of this ancillary lecture is reproduced in the first three supplements. The following supplements (IV–IX) are additions through which the editor has attempted to round off the domain of the material treated in the sense indicated above. Among these, Supplements IV, VI, VIII are essentially only reproductions of known papers by Dirichlet; the others are worked out without such a model, but likewise treat almost exclusively already known subjects. Likewise the final paragraphs (105–110) of the fifth section are added merely for completion. In the preceding sections too there is many a detail that the editor has partly borrowed from older lecture notebooks and partly added according to his own judgment; yet it is not worth the trouble to enumerate everything. The theory of the composition of forms is entirely excluded, because the only paper of Dirichlet's bearing directly on it (*De formarum binariarum secundi gradus compositione*, 1851) treats only the first fundamental theorem, so that the editor had to fear that, in a complete presentation of this theory, he would depart too far from the original purpose of the

whole publication.

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The elaboration of the first sections has deliberately been kept fuller than that of the later ones, in order to lead the beginner gradually more and more to rely on his own powers; and in particular the editor believed that, in the parts added by him, he might be considerably briefer.

LIV. Preface to the second edition of Dirichlet's *Lectures on Number Theory*. 1871

This new edition differs from the first chiefly in that it has been enriched by the tenth supplement, which treats the composition of forms. This subject had remained entirely excluded from the first edition, because the only paper of Dirichlet's that bears directly on it treats only the first fundamental theorem; I therefore had to fear that, in a complete presentation of this theory, I would depart too far from the original purpose of the publication. Although I certainly do not fail to recognize this danger even now, I have nevertheless decided, for many reasons, to add the tenth supplement and thereby, to the best of my ability, to answer repeated requests addressed to me. The main reason is that, despite the uncommon interest and the increasing importance of this theory, no attempt has yet been made to remove the great difficulties that confront the beginner on entering it, while the other sections of the work are excellently suited to facilitate such an attempt. In actually carrying out this decision I did not restrict myself to the foundation of the first elements, but considered it necessary to present, as briefly and simply as possible, the greater part of the investigations contained in the fifth section of the *Disquisitiones Arithmeticae*. Finally, I have included in this supplement a general theory of ideals, in order to cast a new light from a higher standpoint on the main subject of the whole book. Here, to be sure, I have had to restrict myself to presenting the foundations; but I hope that the striving for characteristic fundamental concepts, which in other parts of mathematics has been crowned with such fine success, may not have failed me entirely.

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The investigations in this field created by Kummer, which Kronecker undertook fourteen years ago, have not yet been published; and from that distinguished mathematician's letters at the time I cannot judge in what relations his principles stand to mine. The construction of the theory in § 163 does not yet fully satisfy me; but only after very long reflection did I succeed in giving it this form, whereas about ten years ago, from the theory of higher congruences in connection with the principles of Galois, I had arrived at an entirely different way of founding it, one that has some points of contact with Selling's theory of ideal numbers, but that now seems to me less natural. A more detailed presentation of the algebraic principles connected with the concept of a field (§ 159), which could only be indicated incidentally here, I reserve for another occasion.

It is natural that the addition of the tenth supplement has had some repercussions on the earlier sections. Yet there is no need to fear that, through such changes to the first edition, I have moved farther away in plan and in the posture of the presentation from the real basis, Dirichlet's lectures. To forestall any reproach

of this kind, I repeat here (from the Göttingen learned notices of 27 January 1864) that even the first edition was not based on a notebook written down during the lectures themselves, but only on notes that I had composed from memory and for the most part in extremely brief form. When I heard these lectures as a Privatdozent in Göttingen, I was sufficiently familiar with the material, and my principal aim was to let Dirichlet's extraordinarily penetrating presentation work upon me completely. When the first edition was published, which took place only after a number of years, it became necessary to work these notes out anew and also to supplement them with additions of my own (for example § 2, if I am not mistaken), all of which could not possibly be mentioned. But then as now it has been my most zealous effort to reproduce Dirichlet's presentation with the greatest fidelity. By contrast, I have allowed myself full freedom

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in my own additions. Entirely reworked, for example, are §§ 105 to 110, 143, 144, and some new material has been added partly in the text and partly in the form of notes.

Finally, I have endeavored, wherever possible, to refer to the sources, so as to prompt the reader to study the original works and to awaken in him an image of the progress of science, whose truths, as deep as they are sublime, form a treasure that is the imperishable fruit of a truly noble contest among the peoples of Europe.

Brunswick, 1 March 1871.

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IV. Notice of the second edition of Dirichlet's *Lectures on Number Theory*

[*Göttingische gelehrte Anzeigen*, year 1871, pp. 1481–1494.]

The first edition of this work, which appeared in 1863, was reviewed by me in these pages (27 January 1864), and as regards its origin and content I can in essentials refer to my statements at that time. The new edition differs from the first by a large number of completions, which have been added partly in notes and partly in the text itself. Many paragraphs have also been entirely reworked. These changes, which nevertheless do not touch the essential core of Dirichlet's presentation, have been occasioned mainly by the decision to present, in a new appendix, the tenth supplement, the theory of the composition of binary quadratic forms, which for reasons then mentioned had not been treated in the first edition. The comprehensive generality with which Gauss presented this theory in the fifth section of the *Disquisitiones Arithmeticae* contains considerable difficulties of understanding for the beginner. This circumstance prompted Dirichlet to publish the paper *De formarum binariarum secundi gradus compositione* (1851). He says in its introduction:

De formarum compositione tunc non egi, quod argumentum ab illustrissimo Gauß in "Disquisitionum Arithmeticarum" sectione quinta maxima

quidem generalitate sed per calculos tam prolixos tractatum esse constat, ut perpauca compositionis naturam percipere valuerint eo magis quod summus geometra, ut ipse monuit, brevitati consulens theorematum difficiliorum demonstrationes synthetice adornavit, suppressa analysi per quam erant eruta. Quare confidere posse mihi videor, hujus argumenti expositionem novam et plane elementarem artis analyticae cultoribus non fore ingratham.

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Since in that paper only the first principal theorem of the theory in question is proved, with no indication of its further course, I have taken a somewhat different path, one that agrees with Dirichlet's in considering only a special case of composition. §§ 145–149 contain the general theorems on the composition of forms and form classes. These are used in §§ 150, 151 to find the relation between the class numbers for two determinants that stand to one another as two square numbers; this is the same problem that had already been treated in §§ 97, 99, 100 according to Dirichlet's principles. In §§ 152–154 follow the composition of genera and Gauss's second proof of the reciprocity theorem in the theory of quadratic residues. §§ 155–158 contain a proof of Gauss's theorem that every class of the principal genus arises by duplication; it rests on a theorem of Lagrange and Legendre concerning the solution, in rational numbers, of indeterminate equations of the second degree with two unknowns.

In the paragraphs that then follow I have attempted to introduce the reader into a higher domain in which algebra and number theory are most intimately joined. In the course of the lectures on cyclotomy and higher algebra that I gave in Göttingen in the winter of 1856–1857 before Messrs. Sommer and Bachmann, and in the winter of 1857–1858 before Messrs. Selling and Auwers, I became convinced that the study of the algebraic kinship of numbers is most suitably grounded on a concept that connects immediately with the simplest arithmetical principles. The name “rational domain” that I used then I later replaced by the word “field”; by it I understand a system of infinitely many numbers that has the property that the sums, differences, products, and quotients of any two of these numbers again belong to the same system. I call a field A a divisor of a field M , and M a multiple of A , if all numbers contained in A are also found in M .

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Any two fields A, B always have a least common multiple, which may be denoted by AB , and likewise a greatest common divisor. If to every number a of a field A there corresponds a number $b = \varphi(a)$ in such a way that $\varphi(a + a') = \varphi(a) + \varphi(a')$ and $\varphi(aa') = \varphi(a)\varphi(a')$, then the numbers b form a field $B = \varphi(A)$ conjugate to A , which arises from A by the substitution φ . In the algebraic direction these concepts lead to the principles of Galois; on the number-theoretic side they lead to Kummer's creation of ideal numbers.

In § 159 the most general properties are developed of a field Ω having only a finite number of divisors. In such a field there always exist a finite number of numbers $\omega_1, \omega_2, \dots, \omega_n$ such that every arbitrary number ω of the field can always and in only one way be brought into the form

$$\omega = h_1\omega_1 + h_2\omega_2 + \dots + h_n\omega_n,$$

where $h_1, h_2, \dots, h_n$ denote rational numbers, which I call the coordinates of the

number ω with respect to the basis $\omega_1, \omega_2, \dots, \omega_n$; the number n is called the degree of the field Ω . It then follows easily that every number of the field is an algebraic number, namely the root of an equation of the n th degree whose coefficients are rational numbers, and that the field Ω passes under n different substitutions into n conjugate fields. The product of the n values into which a given number ω of the field passes under these n substitutions is called the norm of ω and is a homogeneous function of the coordinates with rational coefficients, hence a rational number, denoted by $N(\omega)$. Further, if a system of n numbers $\alpha_1, \alpha_2, \dots, \alpha_n$ of the field Ω is given and one forms the determinant from the n^2 corresponding numbers of the n conjugate fields, then its square is a rational number, which I call the discriminant of the numbers $\alpha_1, \alpha_2, \dots, \alpha_n$ and denote by $\Delta(\alpha_1, \alpha_2, \dots, \alpha_n)$. It is not possible, and not necessary, to enter here into the analytic developments attached to these concepts; they too are communicated in this paragraph only to the extent that seemed useful to me for better understanding.

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In the following § 160 all algebraic numbers (which likewise form a field) are divided into whole and fractional numbers. By a whole number is meant every root of an equation whose highest coefficient is $= 1$ and whose remaining coefficients are rational, indeed whole, numbers. From this concept the simplest theorems on divisibility, units, and relatively prime numbers are derived, and these are used later.

The following § 161 contains an auxiliary theorem from a theory by which the concept of congruence of numbers, first introduced by Gauss, is generalized. By a module I understand a system $\mathfrak{m}$ of numbers whose sums and differences belong to the same system; and the congruence $\omega \equiv \omega' \pmod{\mathfrak{m}}$ is to mean that the difference $\omega - \omega'$ is a number of the system $\mathfrak{m}$. This concept has a greater range than its extraordinary simplicity seems to promise; but only what can serve to facilitate the subsequent presentation is communicated here.

After these preparations, in § 162 the whole numbers of a field Ω of degree n are investigated more closely. They form a module $\mathfrak{o}$, and it is first shown that one can always choose n such whole numbers $\omega_1, \dots, \omega_n$ as basis numbers of the field for which every whole number

$$\omega = h_1\omega_1 + h_2\omega_2 + \dots + h_n\omega_n$$

also has whole numbers $h_1, h_2, \dots, h_n$ as coordinates. The discriminant $\Delta(\omega_1, \dots, \omega_n)$ of such a basis, which I call a fundamental system, has, in absolute value, the smallest possible value; and since this rational whole, nonzero number is of especially important significance for the field Ω , it is called the discriminant, or fundamental number, of the field and is denoted by $\Delta(\Omega)$. It divides the discriminant of every system of n whole numbers, and the quotient is a square. Further, if μ is a fixed nonzero number in $\mathfrak{o}$, then the number of numbers contained in $\mathfrak{o}$ and incongruent with respect to μ is equal to the absolute value of the norm $N(\mu)$.

Attention is then drawn to a remarkable phenomenon first observed in the fields arising from cyclotomy. It consists in the fact that a whole number that cannot be further decomposed into a product of whole numbers by no means always plays the role of a true prime number. This was the point of departure for Kummer's creation of ideal numbers.

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I now attempt, in the following § 163, to set up a new theory encompassing all fields; its fundamental idea is as follows. If μ is a nonzero number in $\mathfrak{o}$, then the system $\mathfrak{m}$ of all numbers in $\mathfrak{o}$ divisible by μ has the following two properties:

- I. The sum and the difference of any two numbers in $\mathfrak{m}$ are again numbers in $\mathfrak{m}$; that is, $\mathfrak{m}$ is a module.
- II. Every product of a number in $\mathfrak{m}$ and a number in $\mathfrak{o}$ is again a number in $\mathfrak{m}$.

But one cannot conversely assert that every system $\mathfrak{m}$ of whole numbers of the field that has the two preceding properties, and which from now on I call an ideal, consists of all numbers divisible by some specifiable number μ . If, however, this is the case, then I call $\mathfrak{m}$ a principal ideal and denote it by the symbol $\mathfrak{o}(\mu)$. The properties of all ideals of the field Ω are then investigated, and the following principal result is obtained. If one multiplies each number of an ideal $\mathfrak{a}$ by each number of an ideal $\mathfrak{b}$, then these products and their sums form an ideal, which is called the product of the factors $\mathfrak{a}$ and $\mathfrak{b}$ and is denoted by $\mathfrak{ab}$. According to this definition, $\mathfrak{a}\mathfrak{o} = \mathfrak{a}$, $\mathfrak{ab} = \mathfrak{ba}$, $(\mathfrak{ab})\mathfrak{c} = \mathfrak{a}(\mathfrak{bc})$, and from $\mathfrak{ab} = \mathfrak{ac}$ follows $\mathfrak{b} = \mathfrak{c}$. If an ideal $\mathfrak{p}$ distinct from $\mathfrak{o}$ is called a prime ideal when it has no factor distinct from $\mathfrak{o}$ and $\mathfrak{p}$, then every other composite ideal can always and in only one way be represented as a product of prime ideals. Further, if by the norm $N(\mathfrak{a})$ of an ideal $\mathfrak{a}$ one understands the number of numbers contained in $\mathfrak{o}$ that are incongruent with respect to the module $\mathfrak{a}$, then $N(\mathfrak{ab}) = N(\mathfrak{a})N(\mathfrak{b})$. In this way the complete analogy with the laws of divisibility in rational number theory is established.

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This whole theory is most intimately connected with the so-called theory of higher congruences, which is owed to Gauss's stimulus and to the work of Galois, Schönemann, and others. I was first led by Kummer's papers on the ideal numbers of cyclotomy and by the study of Galois's algebraic investigations to occupy myself in detail with the theory of higher congruences, and at that time I also published a short outline of this theory (Crelle's Journal, vol. 54). Later I tried, with its aid, to establish a general theory of ideal numbers, but was then drawn away from completing it by other work, until the preparations for publishing the present work turned me back to the same subject. The renewed efforts led me to my present theory of ideals, which seems to me to deserve preference over my earlier mode of treatment because it is based on much simpler concepts. In my presentation I have not entered more closely into the connection with the theory of higher congruences, because I feared increasing the size of this appendix all too much. For those readers who wish to occupy themselves more precisely with this connection, I add the following remarks, which may well be useful to them.

Let ω denote an arbitrary number in $\mathfrak{o}$, and put

$$\Delta(1, \omega, \omega^2, \dots, \omega^{n-1}) = D^2 \Delta(\Omega).$$

Then D is always a rational whole number, namely a homogeneous function of the coordinates of degree $\frac{1}{2}n(n-1)$ with rational whole coefficients. If now p is a rational prime number, and if there is a number ω for which D is not divisible by p , then the decomposition of the principal ideal $\mathfrak{o}(p)$ into a product of prime ideals is easily reducible to the theory of higher congruences. Namely, if ω satisfies the equation of degree n , $F(\omega) = 0$, and if

$$F(x) \equiv P_1(x)^{e_1} P_2(x)^{e_2} \dots P_m(x)^{e_m} \pmod{p},$$

where $P_1, P_2, \dots, P_m$ denote distinct prime functions of the variable x , respectively of degrees $f_1, f_2, \dots, f_m$, then

$$\mathfrak{o}(p) = \mathfrak{p}_1^{e_1} \mathfrak{p}_2^{e_2} \cdots \mathfrak{p}_m^{e_m},$$

where $\mathfrak{p}_1, \mathfrak{p}_2, \dots, \mathfrak{p}_m$ denote distinct prime ideals whose norms are respectively $p^{f_1}, p^{f_2}, \dots, p^{f_m}$. From this follows very easily the theorem, extremely fruitful for algebraic and number-theoretic investigations:

Theorem. The prime number p enters the fundamental number $\Delta(\Omega)$ of the field if and only if p is divisible by the square of a prime ideal.

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At first I held it very probable that for every fixed prime number p there would also exist a whole number ω to which would correspond a number D not divisible by p . Only when all my attempts to prove the existence of such a number ω remained fruitless did I set myself the task of demonstrating the incorrectness of this conjecture. If it were correct, then whenever p is divisible by r distinct prime ideals $\mathfrak{p}$ whose norms all have the same value p^f , there would also have to exist at least r distinct prime functions P of degree f ; and conversely, if this latter prerequisite were always fulfilled, then one could also prove the existence of a number ω of the stated character. In the simplest case, when $f = 1$, there are exactly p different prime functions of the first degree. The question therefore is whether there might not exist a field Ω in which p is divisible by at least $(p + 1)$ distinct prime ideals, all having the same norm p ; the degree of the field must then be at least $p + 1$. The simplest case will arise by taking $p = 2$, and one may ask: are there cubic fields in which the number 2 is divisible by three different prime ideals? In such a field D would always be an even number. As a fundamental system of a cubic field one can always choose the number 1 and two other whole numbers α, β whose product is rational. One then has

$$\begin{aligned} \alpha\alpha &= a'\alpha + b\beta - bb', \\ \beta\beta &= a\alpha + b'\beta - aa', \\ \alpha\beta &= ab, \end{aligned}$$

where a, b, a', b' denote rational whole numbers, which in any case have no common divisor, and one finds

$$\Delta(\Omega) = \Delta(1, \alpha, \beta) = a'^2b'^2 + 18aba'b' - 4aa'^3 - 4bb'^3 - 27a^2b^2.$$

If further one sets

$$\omega = z + x\alpha + y\beta,$$

where z, x, y are arbitrary rational whole numbers, then

$$\omega^2 = z^2 - bb'x^2 - aa'y^2 + 2abxy + (a'x^2 + ay^2 + 2xz)\alpha + (bx^2 + b'y^2 + 2yz)\beta,$$

and consequently

$$D = bx^3 - a'x^2y + b'xy^2 - ay^3,$$

independently of z , as was necessary because of the meaning of D . Although a, b, a', b' have no common divisor, D will nevertheless always be an even number if a and b are even and a' and b' are odd. Then the number

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2 must also be divisible by three distinct prime ideals. This is fully confirmed in the example

$$a = b = 2, \quad a' = -b' = 1, \quad \Delta(\Omega) = -503;$$

indeed,

$$\mathfrak{o}(2) = \mathfrak{a}\mathfrak{b}\mathfrak{c}, \quad \mathfrak{o}(\alpha) = \mathfrak{a}^2\mathfrak{c}, \quad \mathfrak{o}(\beta) = \mathfrak{b}^2\mathfrak{c},$$

where $\mathfrak{a}, \mathfrak{b}, \mathfrak{c}$ denote three distinct prime ideals.

Another example is obtained as follows. With respect to the module $p = 2$ there is only a single prime function of the second degree, namely $x^2 + x + 1$; if therefore in a field Ω the number 2 is divisible by at least two distinct prime ideals whose norms are $= p^2 = 4$, then D must always be even. Evidently the degree of the field must be at least 4, and the phenomenon just mentioned does in fact occur in the biquadratic field arising from the equation

$$\alpha^4 - \alpha^3 + \alpha^2 - 2\alpha + 4 = 0;$$

the numbers 1, α , $\beta = 2 : \alpha$, and $\gamma = \alpha^2 - \alpha$ form a fundamental system of it, and its fundamental number is $13^2 \cdot 17$.

Thus there are fields Ω in which all the numbers D are divisible by certain singular prime numbers p , whose number is of course finite. I remark, however, that the general validity of the theorem cited above, by which the character of the rational prime numbers entering the fundamental number $\Delta(\Omega)$ of a field is defined, is in no way lost thereby. It would, however, take us much too far here were I to enter into the proof of this important theorem or into its deeper significance for the kinship of fields. —

After this digression I continue by briefly indicating the content of the following paragraphs. In § 164 all ideals of the field Ω are divided into a finite number of classes. Two ideals are called equivalent if both are transformed into principal ideals by multiplication by one and the same ideal; an ideal class consists of all ideals equivalent to a given ideal; the principal class consists of the principal ideals. These ideal classes then admit a composition in which the same laws prevail as in the composition of classes of quadratic forms.

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In § 165 the connection is proved between the composition of ideal classes and that of decomposable homogeneous forms arising from consideration of the same field Ω .

§ 166 gives Dirichlet's theory of units in a somewhat generalized form that presents itself here quite naturally, and in § 167 it is used to obtain an expression for the number of ideal classes in the form of an infinite series, exactly as in the determination of the number of classes of quadratic forms. At this point, however, I break off the execution of the general problem, since my further investigations in this direction have not yet been crowned with sufficient success to be published. The §§ 168–170 that now follow are intended only to illustrate the preceding general investigations by applying them to the example of quadratic fields.

Up to now the theory of ideal numbers seems to have been the subject of serious research for only four or five mathematicians. It is my heartfelt wish, through the new edition of Dirichlet's *Lectures on Number Theory*, to facilitate access to this great domain and, if possible, to induce a larger number of mathematicians to

devote their powers to it, so that beside the mighty rise that geometry and the theory of functions have undergone in recent times, number theory may not lag behind.

22 July 1871.

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LVI. Notice of P. Bachmann, *The Theory of Cyclotomy and its Relations to Number Theory*

[Literature section of the *Zeitschrift für Mathematik und Physik*, vol. 18, pp. 14–24, 1873.]

I comply all the more gladly with Hofrat Schlömilch's request that I discuss the present work more closely in the *Zeitschrift für Mathematik und Physik*, since I have greeted its appearance with sincere pleasure. A great and highly interesting domain is here, for the first time, made easily accessible to the mathematical learner by an appropriate delimitation and a lucid presentation, and this while presupposing only very moderate preliminary knowledge of algebra and number theory.

The theory of cyclotomy, that is, the theory of equations of the form $x^m = 1$, where m denotes a given positive whole number, had, although many interesting properties of the roots of unity x had already been known earlier, first been brought back to its real principles by Gauss; it thereby became the starting point for an entirely new science of immeasurable extent. The seventh section of the *Disquisitiones Arithmeticae*, published in 1801, which bears the title *De aequationibus circuli sectiones definientibus*, contains a purely algebraic method for reducing the above equation, whose solution by means of trigonometric functions had long been known, to other equations of lower, and indeed of lowest possible, degree. Here the concept of irreducibility appears for the first time (Art. 341), a concept which became decisive for the entire direction of later algebra. Although Gauss makes only slight use of it (Art. 346), I nevertheless do not doubt that this fundamental principle guided him also

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in the discovery of the details, and that he preferred the synthetic presentation only for the sake of brevity. This is suggested especially by the weighty words (Art. 365):

omnique rigore demonstrare possumus, has aequationes elevatas nullo modo nec evitari nec ad inferiores reduci posse, etsi limites hujus operis hanc demonstrationem hic tradere non patiantur, quod tamen monendum esse duximus, ne quis adhuc alias sectiones praeter eas quas theoria nostra suggerit, e.g. in 7, 11, 13, 19 etc. partes, ad constructiones geometricas perducere speret, tempusque inutiliter terat.

The truth of the assertion contained in these words is easy to prove in the present state of algebra, especially since the development and generalization of Gauss's

ideas by Abel and Galois. In fact, from the seed planted by Gauss there has arisen a science which, in order to awaken in a non-specialist at least a dim idea of its character, one might perhaps call the science of the algebraic kinship of numbers, or, to use an expression chosen by me, the science of the kinship of fields. It becomes apparent, namely, that the peculiar constitution of an equation, the possibility of reducing its solution to that of other equations, can be clearly recognized only when, besides its roots, one also considers infinitely many other numbers rationally derivable from one or more of them. The totality of these numbers is precisely what I call a field: a system of numbers which reproduce themselves again and again under the four simplest rational arithmetical operations. On this view, properties of an equation become properties of the corresponding field; relations between equations appear as kinship between fields. In particular, when two fields A , B mutually penetrate one another, there always arises again a field, their least common multiple, or more briefly their product AB , whose nature depends essentially on the kinship of the two fields.

Alongside this development of algebra proper, to which the theory of cyclotomy gave the first impulse, and hand in hand with it, number theory has taken a magnificent upswing.

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Gauss's algebraic investigations in the domain of cyclotomy already required some, though very elementary, auxiliary propositions from number theory; soon, however, it became clear that conversely cyclotomy had become an inexhaustible source from which ever new and significant gains for number theory flowed. One may say that almost all later advances which number theory made in the hands of Gauss, Jacobi, Dirichlet, Eisenstein, Kummer, and Kronecker either owe their very origin to cyclotomy, or, what in some cases was still more remarkable, entered into a previously unsuspected connection with cyclotomy. Among the latter advances are the investigations of Gauss and Dirichlet on the class number of quadratic forms; among the former is Gauss's extension of the concept of a whole number, whose later generalization led to Kummer's creation of ideal numbers.

It is therefore surprising that, despite the role just briefly described which cyclotomy plays in the history of modern mathematics, and despite the great fame which it enjoys before other creations of Gauss at least equally profound — one need only think of the regular heptadecagon —, it is surprising that no textbook has appeared in which cyclotomy is presented as a rounded whole, with full consideration of the details investigated partly by Gauss and partly by his successors. It is therefore a most welcome undertaking of the author to fill this conspicuous gap in our mathematical literature by the present work; and I am pleased to be able to add, for which his name is already sufficient guarantee, that he has carried out this undertaking in excellent fashion. For anyone who intends a deeper study of algebra and its relations to number theory, this work will provide the best guide, since without presupposing extensive prior knowledge it introduces him into the midst of a very rich body of material which has hitherto not been easily accessible, and in which one must be thoroughly oriented if one wishes to proceed to higher investigations.

I now permit myself in what follows to communicate a series of remarks occasioned by individual passages, or also by whole sections, of the work. When I emphasize individual points at which I mention or recommend an arrangement or presentation different from that followed by the author, this is by no means done

in order to censure him.

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Everyone who has occupied himself thoroughly with a definite subject — and for eighteen years I have turned to this part of mathematics with special predilection — forms certain viewpoints peculiar to himself, which he holds to be especially valuable, while they appear less important to another; and I admit that much here is purely a matter of taste. But I gladly use this occasion to make, from my long occupation with this subject, some communications concerning my views, and also some digressions into related matters, in the hope that they will be welcome to some readers. I let them follow here without internal connection, as they arose while reading the work, merely referring to the relevant passage.

Lecture 3, nos. 2 and 3, p. 13. The proof of the existence of primitive roots of unity would, I believe, gain in clarity and precision by a small rearrangement. Since $\varphi(d)$ is defined as the number of d th roots of unity which belong to the exponent d , then, before the contrary has been proved, $\varphi(d)$ appears as depending not only on d , but possibly also on n ; and the applicability of the summation theorem proved in the preceding lecture for determining $\varphi(d)$ remains subject to doubts, which are removed only subsequently by the remarks in no. 3. The investigation is probably simplest if the concept of primitive roots of unity is put first and $\varphi(d)$ is defined as the number of primitive d th roots of unity.

Lecture 4, p. 20. In this lecture the first concepts from the so-called theory of higher congruences are communicated. A somewhat more extensive presentation, also containing the most important propositions on prime functions, would prove very useful on many later occasions, especially for the 5th, 17th, and 18th lectures; but it would of course also require much more space. The proof of Schönemann's theorem (p. 26), assuming Fermat's theorem, can be shortened by

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the theory of the transformation of symmetric functions, which the author does after all presuppose as known in several places, for example in Lecture 5, no. 4.

Lecture 5, nos. 6 and 7. The communication of a special case of Kronecker's general irreducibility theorem leads me to point out here the real nerve of his proof, and also of Arndt's. This can be done with relative brevity if one assumes as known some general concepts concerning whole algebraic numbers and some propositions from the theory of ideals, which I have partly proved in the second edition of Dirichlet's *Lectures on Number Theory*, and partly communicated without proof in the Göttingen *Gelehrte Anzeigen* of 20 September 1871. Let μ denote a primitive m th root of unity; then everything depends on the number-theoretic content of the number $(1 - \mu)$. If $m = 1$, then $1 - \mu = 0$; but if m is divisible by a single prime number p , hence is a power of that prime, then, as is immediately evident,

$$p = \varepsilon(1 - \mu)^{\varphi(m)},$$

where ε denotes a unit and $\varphi(m)$ the familiar function of number theory. Finally, if m is divisible by two or more distinct primes $p, q, \dots$, then the number $(1 - \mu)$, since it divides all numbers of the form $(1 - \mu^n)$, where n denotes any positive whole number, must, by the preceding case, also divide $p, q, \dots$ and consequently be a unit; this can also be concluded directly from the equation whose roots are all primitive m th roots of unity. Now let a be a power of a prime number p , and let $m = ab$, where b is not divisible by p . It is then well known that one obtains all primitive m th roots of

unity, each once only, by multiplying each primitive root of the equation $x^a = 1$ by each primitive root of the equation $x^b = 1$. If α is a definite one of the former and β a definite one of the latter, then Kronecker's theorem, in the following somewhat sharpened form, reads:

If the fundamental number, or discriminant, $\Delta(\Omega)$ of a field Ω is not divisible by the prime number p , then the equation $f(x) = 0$, irreducible in Ω , which is satisfied by $x = \alpha\beta$, also has as roots all $\varphi(a)$ products $\alpha'\beta$ corresponding to all $\varphi(a)$ primitive roots α' of the equation $x^a = 1$.

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The proof rests on the following points. All roots μ of the equation

$$f(x) = \Pi(x - \mu) = 0$$

are in any case of the form $\mu = \alpha'\beta'$, where α' , β' denote primitive roots of unity respectively of degrees a , b . Whenever β' is identical with β , one has $(\beta - \mu) = \beta(1 - \alpha')$, and therefore materially, that is, apart from a unit factor, $= (1 - \alpha)$. If, on the other hand, β' is different from β , then $(\beta - \mu) = \beta(1 - \alpha'\beta'')$, where β'' denotes a root different from 1 of the equation $x^b = 1$; hence, by the preliminary remarks, $(\beta - \mu)$ is a unit. Consequently

$$f(\beta) = \Pi(\beta - \mu) = \varepsilon'(1 - \alpha)^n,$$

where ε' is a unit and n denotes the number of roots $\mu = \alpha'\beta'$ for which $\beta' = \beta$. Therefore $f(\beta)$ will be divisible by

$$p = \varepsilon(1 - \alpha)^{\varphi(a)}$$

always and only when $n = \varphi(a)$, that is, when all $\varphi(a)$ products $\alpha'\beta$ really are roots of the same equation $f(x) = 0$ irreducible in Ω . This divisibility of the number $f(\beta)$ by p can be proved from the assumption that the fundamental number $\Delta(\Omega)$ of the field Ω is not divisible by p as follows. First, this assumption is identical with the assertion that p is divisible by no square of a prime ideal of the field Ω ; and this is in turn equivalent to the assumption that there exist infinitely many powers

$$s = p^f, p^{2f}, p^{3f} \dots$$

of the prime number p for which every whole number ω of the field Ω satisfies the congruence

$$\omega^s \equiv \omega \pmod{p}.$$

Since the coefficients of the function $f(x)$ are such whole numbers ω , it follows that

$$f(\beta)^s \equiv f(\beta^s) \pmod{p}.$$

Choose further, as is always possible, the power s of the prime number p so that $s \equiv 1 \pmod{b}$, hence $\beta^s = \beta$, and at the same time $s \geq \varphi(a)$. Since $f(\beta)$ contains the factor $(1 - \alpha)$ at least once, $f(\beta)^s$ is divisible by p , and it follows that $f(\beta) \equiv 0 \pmod{p}$ as well, as was to be proved.

By repeated application of this result one obtains evidently the following theorem, which is only slightly more general than Kronecker's:

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If m is relatively prime to the fundamental number $\Delta(\Omega)$ of the field Ω , then the equation whose roots are all the primitive m th roots of unity is irreducible in Ω .

Lecture 6, p. 43. In this lecture the proper theory of cyclotomy begins; its essence consists in reducing the equation

$$X = \frac{x^p - 1}{x - 1} = 0,$$

where p denotes a prime number, to equations of lower degree. I permit myself here some general remarks on the distribution of the material and on the method of the investigation.

Gauss's presentation falls into two essentially different parts. The first (Arts. 342–358) contains the successive decomposition of the system of all $(p - 1)$ roots r^k of the above equation into so-called periods, and the setting up of the equations corresponding to them; the second part treats the reduction of these equations to pure equations, that is, their solution by radicals. I consider this sharp separation extremely appropriate. The first part deals only with the numbers rationally expressible by r , which form a field R of degree $(p - 1)$; and it is, even if this does not appear in Gauss's presentation, essentially independent of the existence of primitive congruence roots. The second part, by contrast, requires the consideration of the latter, and also the introduction of an auxiliary field S of degree $\varphi(p - 1)$, which is formed from the roots of the equation $x^{p-1} = 1$. The real object of the investigation is the product RS of the two fields R and S , which at the same time is the field of degree $(p - 1)\varphi(p - 1)$ formed from the roots of the equation $x^{p(p-1)} = 1$. This essential difference between the two parts just named would make it seem to me more advantageous if the author had let the chief content of Lectures 9, 15, 16, 17, 18 follow directly after the seventh.

As to the method of development, it cannot be denied that in Gauss's synthetic presentation the striving for brevity has triumphed over the requirement of deriving everything from a single algebraic idea. During my first thorough study of cyclotomy in the Whitsun vacation of 1855, although I understood the details well, I nevertheless had

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to struggle for a long time before I recognized in irreducibility the principle to which I needed only to put simple, natural questions in order to be driven with necessity to all the details. After these thoughts were completed through a detailed occupation with the algebraic investigations of Abel and especially of Galois, and after the discovery, achieved at the beginning of December of the same year, of the most general relations between any two irreducible equations had brought them to a certain conclusion, I later followed the method then gained in my two winter lectures on cyclotomy and higher algebra in 1856–1858; and I still believe today that it is also appropriate for the learner. Instead of proceeding directly from the periods, as the author does with Gauss, one should ask for the number of the distinct conjugate values which a number $\alpha = F(r)$ contained in R , that is, rationally expressible through r , possesses, and which arise when r is replaced by all $(p - 1)$ roots r^k of the irreducible equation $X = 0$. If h denotes all those f incongruent numbers (mod p) for which $F(r^h) = F(r)$, then irreducibility immediately implies that these numbers h reproduce themselves by multiplication; and from this it follows further that they form all the roots of the congruence $h^f \equiv 1 \pmod{p}$, that

f is a divisor of $(p-1) = ef$, and that e is the number of actually distinct values $F(r^k)$, each of which repeats f times. Namely, all $(p-1)$ exponents k split into e complexes of f exponents hn each, to each of which the one value $F(r^n)$ corresponds. If one asks for the real existence of such e -valued numbers $\alpha = F(r)$, then the condition $F(r) = F(r^h)$ implies that the rational coefficients or coordinates a in the expression $\alpha = F(r) = \sum ar^k$, into which every number contained in R can always, and by irreducibility only in one way, be brought, must be groupwise equal in sets of f ; namely, that

$$\alpha = a\eta + a_1\eta_1 + \cdots + a_{e-1}\eta_{e-1}$$

must hold, where each of the e quantities $\eta, \eta_1, \dots, \eta_{e-1}$ denotes a so-called period, that is, a sum of f terms r^{hn} . If such a number α is actually to be e -valued, only further groupwise equalities among the e rational numbers $a, a_1, \dots, a_{e-1}$

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need be excluded; hence the periods themselves are such e -valued, and indeed conjugate, numbers. It follows at once that every e -valued number α is the root of an irreducible equation of degree e , and that every number $\beta = f(r)$ satisfying the conditions $f(r) = f(r^h)$, even without being e -valued, is rationally expressible through α . Thus all such numbers β form a field R_e of degree e , completely determined by e . The most general result, in which the whole first part of cyclotomy is contained and which is easily derived in the same way, is the following. If α is an e -valued number, α' an e' -valued number, and e'' is the greatest common divisor of e and $e' = d'e''$, then α' satisfies an equation irreducible in R_e of degree d' . The same can also be expressed as follows: the product of the fields R_e and $R_{e'}$ is the field $R_{e'''}$, where e''' denotes the least common multiple of e, e' ; and $R_{e''}$ is their greatest common divisor. This forms only a special case of the general law mentioned above, which can be expressed at least partially by the equation

$$(BC, A) = (C, AB)(B, A),$$

where A, B, C are any three fields, and the symbol (B, A) denotes the number of numbers contained in B which are independent with respect to A .

In the second part of cyclotomy the question is raised whether the irreducible equations considered in the first part are soluble by radicals, that is, reducible to pure equations. This question necessarily presses toward the introduction of the field S , and once the irreducibility of the equation $X = 0$ with respect to S has been recognized, it is immediately clear that the propositions of the first part can be transferred directly to the new domain. If one now asks for such numbers as satisfy pure equations, then one is driven in the same way to all the resolvents and to the use of primitive congruence roots, just as in the first part one was driven to the periods; and all the detail follows with the greatest ease.

The path described above is followed by the author only in part; but I believe that clarifying the algebraic principles in the classical and at the same time simplest example of cyclotomy is a very useful preparation for the deeper study of algebra.

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Lecture 9, no. 4. Kronecker's method for determining the sign $\varepsilon = \pm 1$ in the equation

$$\varepsilon \Pi(x^{2h-1} - x^{p-2h+1}) = \sum \left(\frac{k}{p} \right) x^k + (x^p - 1)f(x)$$

can be shortened by dividing by $(x-1)^{\frac{p-1}{2}}$ and then putting $x = 1$.

Lecture 14. After the reciprocity laws for cubic and biquadratic residues have been proved in the theories of complex numbers, it is of interest to return to the theory of rational numbers. Although this subject by its nature does not belong to the present work, I nevertheless permit myself to communicate here the following from a larger work on arbitrary cubic fields, whose completion and publication I am for the present prevented from undertaking by official duties. Let k denote a whole rational number whose cube root is irrational; then from the equation $x^3 = k$ there arises a pure cubic field whose fundamental number has the form $D = -3g^2$, where g is a whole number easily derived from k . If one now asks for all prime numbers p not dividing k , of the form $3n+1$, for which the given number k is a cubic residue, then by means of the reciprocity law one reaches the following interesting result, which was in essence already known to Gauss: all non-equivalent, primitive positive quadratic forms $ax^2 + bxy + cy^2$ in which $b^2 - 4ac = D$ split into three divisions with equally many individuals, the first of which forms a group. By the forms of this first division all and only those prime numbers p are represented for which k is a cubic residue. With its help, the determination of the number of ideal classes of the cubic field is reduced to a known part of the theory of theta functions.

Lectures 17 and 18. In founding the theory of ideal numbers in cyclotomy, the author follows the path taken by Kummer. The attempt to overcome some doubts about the rigor of Kummer's first presentation, later completed, first led me in the winter of 1855/56 to the theory of higher congruences, in particular to the decomposition of the function $(x^m - 1)$ into prime functions with respect to arbitrary prime moduli. These propositions, which incidentally, if I am not mistaken,

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were first published by Schönemann and have also been found in Gauss's Nachlass, allow, it seems to me, a simpler foundation of Kummer's ingenious theory. Retain the former meaning of X , and let q be a prime different from p which belongs (mod p) to the divisor f of $(p - 1) = ef$. Then

$$X \equiv Q_0 Q_1 \cdots Q_{e-1} \pmod{q},$$

where the e functions Q denote prime functions of degree f . Let h again denote all the f numbers $1, q, q^2, \dots, q^{f-1}$ incongruent with respect to p , which form the roots of the congruence $h^f \equiv 1 \pmod{p}$, and let Q, Q' be any two of the e prime functions. Then the congruence

$$Q(y) \equiv 0 \pmod{q, Q'}$$

has each time f incongruent roots $y = x^{hn}$. If one now writes the e functions in the form

$$Q = x^f - ux^{f-1} + \cdots \pmod{q},$$

then it follows incidentally that

$$\Pi(x - \eta) = F(x) \equiv \Pi(x - u) \pmod{q},$$

where $F(x)$ denotes the function of degree e which vanishes for the e periods η of f terms. But this is by no means required for the further foundation. The e numbers which the author denotes by ψ can, in effect, be completely replaced by the e numbers which arise from the product

$$Q_0(r)Q_1(r) \cdots Q_{e-1}(r) = qf(r)$$

by omitting each time one of the e factors $Q(r)$. The proof, especially in the theorems on powers of the ideal prime numbers, is simplified if, as is always possible,

the e functions Q are chosen so that $f(r)$ is relatively prime to q , that is, so that $f(x)$ becomes a relative prime function to $X \pmod{q}$. Instead of this, however, one may also use the decomposition of the function X into e factors with respect to arbitrarily high powers of q . I note here that the author's definition (on p. 262) of the powers of ideal prime numbers leaves a small and easily removed doubt, insofar as it is not proved that a number which contains a prime factor m times must certainly contain it also $(m-1)$ times. Until this is done, it remains conceivable, for example, that a number contains a prime factor exactly six times and at the same time exactly eight times.

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A similar doubtful place is also found in the general theory of ideals which I presented in the second edition of Dirichlet's *Lectures on Number Theory*: the assertion that the concept of powers of a prime ideal is wholly independent of the accidental definition of the latter (§ 163, 4) is not properly justified, but it can easily be proved from the propositions immediately following it. The whole presentation can be considerably simplified by some rearrangements, and an even greater simplification will presumably succeed for those who approach the material without preconceptions and with fresh powers. Still, I believe that by this theory, whose construction cost me several years of indescribable exertion, I have achieved something useful. I am, to be sure, far from wanting to compare its value with that of Kummer's first creative idea; but apart from the aesthetic satisfaction afforded by the recognition that the same simple laws according to which, in rational number theory, numbers are composed from prime numbers recur in the theory of ideals of every field, the certainty of the general theory has the great practical advantage that in each special case one need not again and again seek out the first foundations, and that the discovery of the special laws is made extraordinarily easier. How great this benefit is may be seen quite clearly in the present case of cyclotomy: the nature of all ideals occurring here follows — as one will easily convince oneself — with a few strokes of the pen from the single remark that the p numbers $1, r, r^2, \dots, r^{p-1}$ are certainly incongruent with respect to every prime ideal not dividing p , because their differences all divide the number p .

I close my already too extensive remarks on the present work by once more most warmly recommending it for study to all who wish to acquire a thorough and many-sided knowledge of cyclotomy and its relations to number theory, and by expressing to the honored author my thanks for publishing these lectures of his.

Brunswick, 1 February 1873.

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Explanatory notes to the preceding article

Kronecker's original theorem on the irreducibility of the cyclotomic equations says that the equation $\varphi_n(x) = 0$ of the primitive n th roots of unity is irreducible in an algebraic field $K(a)$ if the equation discriminant of a is relatively prime to n . In the remarks on Lecture 5, however, Dedekind proves something sharper: that it already suffices if the field discriminant of $K(a)$ is relatively prime to n .

The investigations concerning the class number of cubic fields which Dedekind mentions in his remarks on Lecture 14 were later published by him in the paper “Über die Anzahl der Idealklassen in reinen kubischen Zahlkörpern” (XXIX); compare the introduction to that work.

Ore.

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LVII. Notice of the first edition of Riemann’s collected works

[*Göttingische gelehrte Anzeigen*, year 1876, pp. 961–965.]

If, in consequence of a request addressed to me, I permit myself to announce the appearance of this work in these pages, I do so not with the intention of entering more closely into its contents. For such a notice has already been composed, more perfectly than I could manage, by the actual editor, my highly honored friend Heinrich Weber in Königsberg, and will soon be published elsewhere, in the repertory of L. Königsberger and G. Zeuner. The purpose of the following lines is rather only to make some communications concerning the origin of the edition, which may perhaps be of some interest to the readers of the *G. G. A.*

Soon after Riemann’s death I received from Frau Professor Riemann the honorable commission to examine the scientific Nachlass and to publish what was suitable for publication. Three papers, namely those on trigonometric series, on the hypotheses of geometry, and the contribution to electrodynamics, I found completed in the cleanest handwriting, and I hastened to publish them. The rest, with few exceptions, was in an entirely unordered state, easily explained by Riemann’s many journeys; and before all else it was necessary to examine, according to their contents, the extraordinary number of separate leaves that had accumulated since Riemann’s student days, to recognize what belonged together, and to order it. Only someone who has looked into these papers can form a clear conception of the nature of this preparatory activity. For the most part they contain,

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in many repetitions, drafts for the papers published by Riemann himself and preparations for his lectures. In addition there are excerpts from the most varied works, copies of passages that had especially aroused Riemann’s interest, and drafts of letters. Many leaves are covered with formulas without any explanatory text, and all these things are sometimes united on one and the same folio sheet, written partly from top to bottom, partly in the opposite direction. Under these circumstances my efforts to order the papers met with only imperfect success; and although I had worked through Riemann’s works again and again with zeal, I could not conceal from myself that for deciphering the content of some papers I did not possess the full detail knowledge required, since my own studies as a whole were directed toward another domain of mathematics. I therefore at first contented myself with putting together in fair copies what was intelligible to me, and intended to proceed to publish some of it, especially the Paris prize memoir together with a commentary on its relations to the paper on the hypotheses of

geometry, when I was disturbed in my plan for several years by the necessary preliminary work for a second edition of Dirichlet's number theory. Soon after I had turned back to it, a new idea arose in Göttingen: to publish Riemann's works collected in full. Clebsch, Riemann's successor, probably stimulated by Wilhelm Weber, had seized on this idea with all his liveliness in the spring of 1872; and when he visited me at Whitsuntide I gladly entered into his plan to include the Nachlass as well in this edition. At my request Clebsch took over the chief direction of the undertaking and had all the papers sent to Göttingen in order to subject them to another inspection and examination, which I considered urgently necessary for the reasons given above. I myself, since soon afterwards a worrisome secondary office absorbing all my strength was entrusted to me for three years, could take part in the intended edition only to a small degree. It promised to proceed rapidly, and printing was soon to begin, when the sudden and deeply lamentable passing of Clebsch in November

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1872 brought the enterprise completely to a halt. After for some time I had received no news at all of its progress, I tried to persuade an outstanding mathematician and friend of Clebsch to take his place; unfortunately he was prevented by the most compelling reasons from fulfilling my request. Thus the matter again lay idle, since I myself, because of my duties, was unable to take up the edition. With Frau Professor Riemann's consent I finally resolved, in November 1874, to ask Professor H. Weber in Zürich, who by his works had proved himself one of the deepest connoisseurs of Riemann's creations, to take the great work entirely into his own hands; and the smaller my hope had been of seeing my request fulfilled, the greater was my joy when, despite serious reservations, he declared himself unconditionally ready to devote his powers wholly to this undertaking. From that moment the matter took the most gratifying and rapid course. Besides the careful revision of the papers already published, it was a matter of another exact examination of the entire Nachlass. This laborious work yielded, besides other successes whose enumeration here would lead too far, the very fortunate result that several components of the Nachlass whose meaning had escaped me, or whose value I had not fully appreciated, could be included in the edition or at least used for it. It is therefore solely Professor Weber's merit that the work could so soon, and in such completeness, be handed over to the mathematical public. My own collaboration, apart from a few trifles, was thereafter limited to participation in the revision of the proofs.

Brunswick.

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LVIII. Preface to the third edition of Dirichlet's *Lectures on Number Theory*. 1879

Concerning the origin of the present work, the first half of which essentially reproduces a lecture given by Dirichlet in Göttingen in the winter of 1856 to 1857, I have given the necessary report in the prefaces to the first two editions and in the Göttingen *Gelehrte Anzeigen* of 27 January 1864 and 20 September 1871. The

principal purpose of the edition, to introduce the beginner to the elements of number theory and especially to the theory of binary quadratic forms by transmitting as faithfully as possible Dirichlet's perfected lecture, has still been retained; and I have not been able to decide to make any substantial alteration in this part of the work. The additions that I had already inserted in the earlier editions for completion, which begin with the fourth supplement and are partly worked out from papers by Dirichlet and partly arose from my own investigations, I have again considerably increased. Especially to be mentioned is the broader presentation, contained in the final supplement, of the same ideal theory which I first published in the second edition, but in so compressed a form that the wish for a more detailed treatment was expressed to me from several sides. I have complied with this request all the more gladly since an investigation carried out jointly by my friend H. Weber in Königsberg and myself, and soon to appear, has yielded the result that the same principles can be successfully transferred to the theory of algebraic functions. The new presentation, in which much is borrowed verbatim from the work *Sur la théorie des nombres entiers algébriques* published three years ago,

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has of course required much more space; but this will probably find excuse if, as I hope, one is convinced that the unified character of the whole work has by no means suffered. I also use this opportunity to draw the reader's attention to the papers by Selling (in the tenth volume of Schlömilch's *Zeitschrift für Mathematik und Physik*, 1865) and by Zolotareff (in the sixth volume of the third series of the journal of Liouville and Resal, 1880), in which the theory of ideals is founded on that of higher congruences; and finally I may express the hope that Kronecker's relevant investigations, which stem from an earlier period, will soon be published.

Brunswick, 11 November 1880.

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LIX. Preface to the fourth edition of Dirichlet's *Lectures on Number Theory*. 1894

In the prefaces to the first three editions (1863, 1871, 1879–1880) and in the Göttingen *Gelehrte Anzeigen* of 27 January 1864 and 20 September 1871, I have made the necessary communications concerning the first origin and gradual expansion of this work; I refer to them here. Even now, in publishing the fourth edition, the first part, which essentially reproduces a lecture delivered by Dirichlet in Göttingen in the winter of 1856 to 1857 and extends to § 120, has remained almost unchanged; and I have contented myself with adding a few new remarks to my earlier notes in order occasionally to point out another method of proof or also more recent appearances in the literature. The same is true of my additions, which are partly worked out from papers by Dirichlet and partly arose from my own investigations. Only the final supplement, which treats the general theory of whole algebraic numbers, has undergone a complete revision; both the algebraic foundations and the properly number-theoretic foundations are presented in the greatest detail and from the point of view which, after many years of conviction, I consider the simplest, because it presupposes chiefly only a clear survey of the realm of

numbers and knowledge of the rational fundamental operations. The same conception also underlies some works on ideal theory which I hope to publish soon; and so it may be excused if I have treated some things more thoroughly than the immediate aims of the present work seem to require.

In the great Festschrift *Grundzüge einer arithmetischen Theorie der algebraischen Größen*, by which Kronecker in advance erected a beautiful monument to his teacher and friend Kummer, who passed away so soon after him, there are laid down the comprehensive ideas upon which another theory of ideals rises. By publishing it (1882 in Crelle's

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Journal, vol. 92), Kronecker fulfilled a wish which I had often expressed, most recently in June 1880 on the occasion of the unveiling of our Brunswick statue of Gauss, when it was also agreed that this paper should appear in Crelle's Journal before the *Theorie der algebraischen Funktionen einer Veränderlichen* worked out by H. Weber and me. Its content was also completely new to me, since on the basis of an old letter communication from 1857 I had believed that Kronecker's theory had to be sought along entirely different paths, which I indicated in § 10 of my work *Sur la théorie des nombres entiers algébriques* (1877). I do not yet consider it possible to pronounce a secure judgment on the advantages and disadvantages of this theory, whose great significance is beyond doubt, since in the paper only the fundamental ideas are sketched in broad strokes; I should therefore like to express the wish that one of Kronecker's numerous pupils undertake to work out a complete, systematic presentation of this theory. I recently tried to give a small contribution to this in the *Mitteilungen der Deutschen mathematischen Gesellschaft in Prag* (1892). If one considers what transformations other parts of mathematics, for example the theory of elliptic functions, have undergone over time since their first beginnings, one will find it very probable that still simpler foundations than those known so far will also be found for ideal theory. As such a foundation one may consider, for example, the theorem derived by me from ideal theory (pp. 465, 541, 577 of the second, third, and fourth editions of this work) concerning the greatest common divisor of two arbitrary whole algebraic numbers. I tried many years ago to take this path; but I did not succeed in obtaining an essential simplification, because I could give the direct proof of this theorem only with the same means that in essence underlie my theory of ideals as well. Nevertheless, I would recommend this path to the attention of younger mathematicians who enter this field of research without preconceptions, and to whom such a proof may therefore more easily succeed than it did to me.

Bad Harzburg, 30 September 1893.